

2023. 12:47.

Q1. (a). $G(s) = \frac{C(s)}{R(s)} = \frac{k_p \cdot \frac{2}{s(s^2+4s+2)}}{1 + k_p \cdot \frac{2}{s(s^2+4s+2)}} = \frac{2k_p}{s^3+4s^2+2s+2k_p}$

From Routh Hurwitz's stability criterion

s^3	1	2		$b_1 = \frac{1}{-4} \left \begin{matrix} 1 & 2 \\ 4 & 2k_p \end{matrix} \right = -\frac{1}{4} (2k_p - 8) = \frac{4-k_p}{2}$
s^2	4	$2k_p$		
s^1	$\frac{4-k_p}{2}$	0		$b_2 = 0$
s^0	$2k_p$			$c_1 = -\frac{1}{b_1} \left \begin{matrix} 4 & 2k_p \\ b_1 & 0 \end{matrix} \right = -\frac{1}{b_1} (-2k_p b_1) = 2k_p$

For this system to be stable: $\begin{cases} \frac{4-k_p}{2} > 0 \\ 2k_p > 0 \end{cases} \Rightarrow \begin{cases} k_p < 4 \\ k_p > 0 \end{cases}$

$\Rightarrow 0 < k_p < 4$

(b). For a unit impulse input: $R(s) = 1$

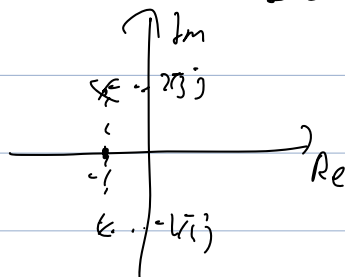
$C(s) = G(s) \cdot R(s) = \frac{s+1}{s^2+5s+6} = \frac{s+1}{(s+2)(s+3)} = -\frac{1}{s+2} + \frac{2}{s+3}$

$\Rightarrow c(t) = -e^{-2t} + 2e^{-3t}$

(c). $\frac{C(s)}{R(s)} = \frac{(s+1) \cdot \frac{1}{s^2+5s+6}}{1 + (s+1) \cdot \frac{1}{s^2+5s+6}} = \frac{s+1}{s^2+2s+3}$

This system has 2 poles: $p_1 = -1 + 2\sqrt{3}j$ $p_2 = -1 - 2\sqrt{3}j$

1 zero: $z_1 = -1$



Stable

Oscillate with decreasing amplitude

$\frac{k_p s + 4}{s^2 + 5s + 6}$

$$(d). \frac{K_D}{K_D} = \frac{1}{1 + \frac{K_D + s K_D}{s^2 + s + 8}} = \frac{K_D + s K_D}{s^2 + (1 + K_D)s + 8 + K_D}$$

Its characteristic equation is $s^2 + (1 + K_D)s + 8 + K_D = 0$

$$\text{compare with } s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\Rightarrow 2\zeta\omega_n = 1 + K_D \quad \omega_n^2 = 8 + K_D$$

$$\Rightarrow \left\{ \begin{array}{l} K_D = \omega_n^2 - 8 = 2.89 \\ K_D = 2\zeta\omega_n - 1 = 1 \end{array} \right.$$

$$f_s = \frac{3}{\zeta\omega_n} = 3 \quad \zeta\omega_n = 1$$

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = 1 \quad 1 - \zeta^2 = \zeta^2$$

$$\frac{\sqrt{1 - \zeta^2}}{\zeta} = \pi \Rightarrow \zeta = 0.303$$

$$\omega_n = 3.300$$

Q2.

$$(a). \text{ If } G_c(s) = K_D$$

$$\frac{h(s)}{h_R(s)} = \frac{\frac{K_D}{s^2}}{1 + \frac{K_D}{s^2}} = \frac{K_D}{s^2 + K_D}$$

$$\text{When } h_R(s) = \frac{1}{s}$$

$$h(s) = \frac{K_D}{s(s^2 + K_D)} = \frac{A}{s} + \frac{Bs + C}{s^2 + K_D} = \frac{A(s^2 + K_D) + (Bs + C)s}{s(s^2 + K_D)}$$

$$= \frac{(A + B)s + Cs + AK_D}{s(s^2 + K_D)}$$

$$\Rightarrow \left\{ \begin{array}{l} A + B = 0 \\ C = 0 \\ AK_D = K_D \end{array} \right.$$

$$\Rightarrow \left\{ \begin{array}{l} A = 1 \\ B = -1 \\ C = 0 \end{array} \right. \Rightarrow h(s) = \frac{1}{s} - \frac{1}{s^2 + K_D}$$

$$\Rightarrow h(t) = 1 - \cos\sqrt{K_D} t$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{K_D}}$$

$$(b). \frac{h(s)}{h_R(s)} = \frac{\frac{K_d + K_d s}{s^2}}{1 + \frac{K_p + K_d s}{s^2}} = \frac{K_d s + K_p}{s^2 + K_d s + K_p}$$

to make it non-oscillatory: $K_d^2 \geq 4K_p$

$$(c). \quad K_d = 2\zeta\omega_n \quad \Rightarrow \quad \begin{cases} \omega_n = \sqrt{K_p} = 4 \text{ rad/s.} \\ \zeta = 1 \end{cases}$$

$$K_p = \omega_n^2$$

$$64 \geq 4 \times 16$$

it satisfies $K_d^2 \geq 4K_p$, non-...

$$\begin{aligned} h(s) &= \frac{1}{s} \cdot \frac{8s + 16}{s^2 + 8s + 16} = \frac{A}{s} + \frac{Bs + C}{(s+4)^2} + \frac{D}{s+4} \\ &= \frac{A(s+4)^2 + (Bs+C)(s+4) + D \cdot s}{s(s+4)^2} \\ &= \frac{(A+B+D)s^2 + (8A+C+4D)s + 16A}{s(s+4)^2} \end{aligned}$$

$$\Rightarrow \begin{cases} A+B+D=0 \\ 8A+C+4D=8 \\ 16A=16 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=0 \\ C=4 \\ D=-1 \end{cases} \quad \text{(Let } B=0)$$

$$\Rightarrow h(s) = \frac{1}{s} + \frac{4}{(s+4)^2} - \frac{1}{s+4}$$

$$\Rightarrow h(t) = 1 + 4e^{-4t} \cdot t - e^{-4t}$$

QB:

$$(a), (i), G_0(s) = \frac{K(s+6)}{(s+3)(s+5)(s+10)}$$

3 poles: $-3, -5, -10$

1 zero, -6

(ii), 2

$$\chi = \frac{-3-5-10+6}{2} = -6$$

$$\frac{dG(s)}{ds} = \frac{s^3 + 18s^2 + 95s + 150 - (s+6)(3s^2 + 36s + 95)}{(s^3 + 18s^2 + 95s + 150)^2}$$

$$= \frac{-2s^3 - 36s^2 - 216s - 420}{()^2}$$

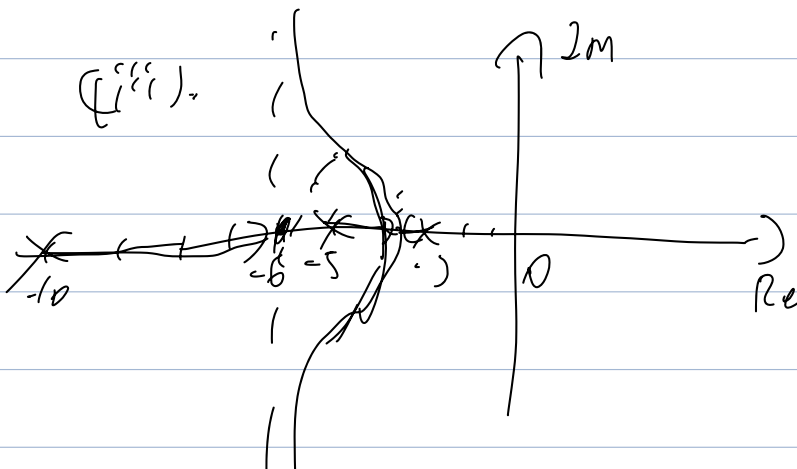
$$3s^3 + 36s^2 + 95s + 18s^2 + 216s + 570$$

$$3s^3 + 52s^2 + 311s + 570$$

$$s_1 = -4.18$$

$$s_2 = \dots$$

$$s_3 = \dots$$



(iv), for all $K > 0$ stable. Oscillates with decaying comp.

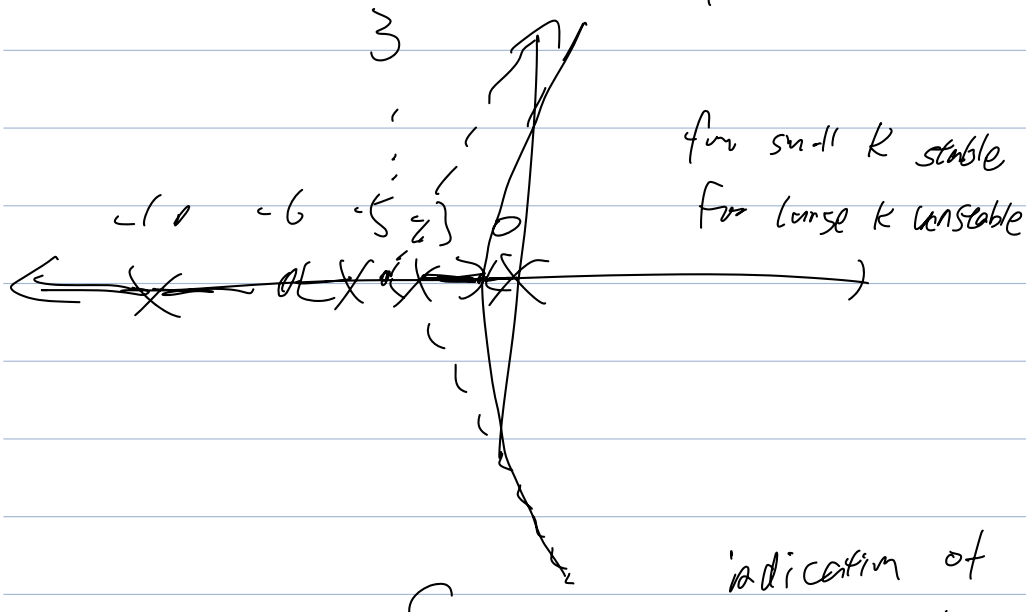
nv.

$$(v), \frac{s+6}{(s+3)(s+5)(s+10)} = -\frac{1}{K} \Rightarrow K = 3.09$$

$$(vi), \frac{K}{s} \cdot \frac{s+6}{(s+3)(s+5)(s+10)} \quad -18+6$$

$4 - 1 = 3$ asymptotes

$$\frac{0 - (-5 - 10 - (-6))}{3} = -4$$



indication of
relative stability

